

RAG Deep Dive (Retrieval Augmented Generation)

The Most Important Topic for an AI Solution Architect

 If someone asks:

What is the most important concept in Enterprise Generative AI?

The answer is:

RAG (Retrieval Augmented Generation)

Most enterprise AI solutions today are built using:

- Enterprise Search
- Knowledge Assistants
- Employee Assistants
- Support Bots
- AI Copilots
- AI Agents

All of these depend on RAG.

Why RAG Exists

Imagine a user asks:

What is the IDD Tech PR approval process?

GPT cannot answer.

Why?

Because GPT was not trained on:

IDD Tech Documents

Project Runbooks

Support Guides

Architecture Documents

Internal Policies

Traditional GPT Problem

User Question

↓

GPT

↓

Answer

Problems:

✗ No company knowledge

✗ No access to documents

✗ Hallucinations

✗ Outdated answers

Solution

Retrieve company information before asking GPT.

Retrieval

Augmented

Generation

What is RAG?

RAG has three parts.

Retrieval

Find relevant information.

Example:

Search Project Documentation

Augmentation

Attach documents to the prompt.

Example:

User Question

+

Relevant Documents

Generation

Generate final answer.

GPT now uses:

User Question

+

Retrieved Content

Simple RAG Architecture

User Question



Search Documents



Relevant Content



GPT



Answer

Traditional GPT vs RAG

Traditional GPT

Question



GPT



Answer

Problem:

No Internal Knowledge

RAG

Question



Document Search



Context Retrieval



GPT



Answer

Benefits:

- ✓ Company Knowledge
 - ✓ Better Accuracy
 - ✓ Current Information
 - ✓ Reduced Hallucinations
-

Complete Enterprise RAG Architecture

Documents



Chunking



Embeddings



Azure AI Search



User Question



Question Embedding



Vector Search



Relevant Chunks



Azure OpenAI



Answer

Topic 1 - Chunking

One of the most important concepts.

Problem

Document:

Employee Handbook

500 Pages

Cannot send entire document.

Solution

Split document.

Chunk 1

Chunk 2

Chunk 3

Chunk 4

Example

Document:

Travel Policy

becomes:

Chunk 1

Introduction

Chunk 2

Travel Requests

Chunk 3

Approvals

Chunk 4

Reimbursements

Why Chunking?

Benefits:

- ✓ Faster
 - ✓ Lower Cost
 - ✓ Better Retrieval
 - ✓ Better Answers
 - ✓ Better Search Accuracy
-

Topic 2 - Embedding Pipeline

Documents become vectors.

Document

↓

Chunking

↓

Embedding Model

↓

Vector

↓

Vector Index

Example

Project Document



Chunking



Azure OpenAI Embeddings



Vectors



Azure AI Search

Topic 3 - Retrieval

User asks:

How does leave approval work?

System performs:

Question



Embedding



Vector Search



Relevant Chunks

Output:

Leave Policy Section

Topic 4 - Context Injection

Retrieved content is injected into GPT.

Example:

Context:

Leave approval requires manager approval.

Question:

How does leave approval work?

GPT answer:

Based on retrieved content,
manager approval is required.

Topic 5 - Grounding

Grounding means:

Only answer from retrieved content.

Without Grounding:

GPT guesses.

With Grounding:

GPT uses actual documents.

Benefits:

- ✓ Accuracy
 - ✓ Reliability
 - ✓ Trust
-

Topic 6 - Hallucinations

Hallucination means:

AI confidently generates
incorrect information.

Example:

Question:

What is our reimbursement policy?

Without RAG:

GPT may invent an answer.

With RAG:

GPT reads company policy.

Result:

✓ More Accurate

✓ Less Hallucination

Topic 7 - Enterprise RAG Architecture

SharePoint

Blob Storage

SQL

PDF Files

Word Documents

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Document Processing

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Chunking

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Azure OpenAI Embeddings

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Azure AI Search

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Retriever API (.NET)

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Azure OpenAI GPT

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Application

Architect View

A developer asks:

What is RAG?

A Solution Architect asks:

Where does data come from?

How will documents be chunked?

Where will embeddings be stored?

How will access be controlled?

How is monitoring implemented?

How is cost optimized?

How is security enforced?

Security Architecture

Always include:

Authentication

Microsoft Entra ID

Authorization

Azure RBAC

Secrets

Azure Key Vault

Monitoring

Azure Monitor

Application Insights

Network Security

Private Endpoint

Cost Optimization

Avoid:

Sending entire 100-page documents
to GPT

Use:

Chunking

Benefits:

- ✓ Lower Token Usage
 - ✓ Faster Responses
 - ✓ Lower Cost
 - ✓ Better Accuracy
-

RAG vs Fine Tuning

RAG

Uses:

External Data

Best For:

- Document Search
 - Knowledge Assistants
 - Internal Policies
 - Enterprise Search
-

Fine Tuning

Changes:

Model Behavior

Best For:

- Specialized Output Style
- Domain-Specific Writing Style

Enterprise Recommendation

Most customers should start with:

✅ RAG

Before considering:

❌ Fine Tuning